Project Proposal

Submitted.By:

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Subject:

Information Security

Project Title:

AI-Based Adaptive Login Defense System Using Agentic Security Principles.

1. Project Idea:

Most login systems today use fixed security rules, like simple password checking or showing a CAPTCHA after multiple failed attempts. These static systems are not smart; they cannot adjust to different types of attacks or user behavior.

This project proposes an **agent-based adaptive login system** that can monitor user activity and respond intelligently to suspicious actions. The system will analyze login attempts, calculate a dynamic risk score, and decide the best defensive action automatically.

The main goal is to simulate an intelligent security agent that protects user accounts from brute-force and automated login attacks by adapting its response based on the perceived threat.

2. Scope:

This project will create a console-based login system that:

- Monitors login attempts and user behavior
- Detects unusual patterns, such as repeated failures or very fast attempts
- Calculates a dynamic risk score for each user
- Automatically triggers appropriate security actions, like:
    - Displaying a CAPTCHA
    - Adding a delay between login attempts
    - Temporarily locking the account

The project will simulate automated attack behavior and show how an adaptive system can respond differently based on risk level.

This system will focus on agentic decision-making in cybersecurity. It will not include real-time web deployment or advanced AI models but will demonstrate intelligent behavior using rule-based logic and dynamic responses.